

Syllabus for Physics Courses

PHYS101 General Physics (1):

2 Credit hrs. Lecture + 1 Credit hr. Practical

Prerequisites Courses: -

Dimensions and Physical units - Vectors - Static and dynamic of solid bodies - Conservation laws of energy and motion - general gravity – Elasticity – Viscosity - Surface tension - Capillary tube - Different methods for determination of surface tension - Bernoulli's equation - Poisson's equation - Stock law - Concept of heat - Adiabatic variation - Thermometers - Heat expansion - Isothermal variation - Heat capacity - Specific heat - Gases Liquefaction - Humidity - Transfer of heat - Radiation and its application - Law of gases.

References:

- Loney, S. Humphery, H. Serway.

PHYS102 General Physics (2):

2 Credit hrs. Lecture + 1 Credit hr. Practical

Prerequisites Courses: PHYS101

Electric charge - Coulomb's law - The electric field - Lines of electric forces - Gauss's law - Electric potential difference - The relation between potential difference and electric field - Capacitance and Capacitors - Isolators - Properties of dielectrics - Electric current and resistance - Power and Energy - Kirshoff's law - Whetston's Bridge - Metric Bridge - Electrical potential and potentiometer - Magnetization - Inverse square law - Moment of inertia - The effect of two perpendicular fields - Laplace's equation - Magnetic field Strength - Measurement apparatus - Law of motion - Rotational motion - Vibration motion - Simple harmonic motion - Waves - Sound and light waves - Reflection and refraction laws - Mirrors and lenses - Prism - Spectra - Interfaces and diffraction - Hygen's theory.

References:

- Loney, S. Humphery, H. Serway.

PHYS141 Classical mechanics (1):

2 Credit hrs. Lecture + 1 Credit hr. Practical

Prerequisites Courses: -

Vector algebra - Force and linear motion - motion in plane (projectiles - Circular motion - Polar coordinates) - Work and energy - linear momentum and circular momentum - impulse rocket - elastic and inelastic collision.

PHYS142 Classical mechanics (2):

2 Credit hrs. Lecture + 1 Credit hr. Practical

Prerequisites Courses: PHYS141

Motion in direction of central force- potential energy- motion equation- Gravitational force and kepler law- system of particle- circular motion- moment of inertia- Theory of parallel axes- Rotational dynamic of particle- Application on simple and compound pendulum- motion resistance- Fraction of motion.

PHYS211 Physical Optics:

2 Credit hrs. Lecture + 1 Credit hr. Practical

Prerequisites Courses: PHYS102

Wave motion – Superposition of wave – Young's experimental – Relation between the amplitude of reflected and refracted beam at plane surface – Interference involving multiple reflection – Reflection from plane parallel film - Reflection from film of varying thickness – Newton rings – Interferometers and its applications – Fresnel's diffraction - Fresnel's half - period zone – Zone plate- Luminosity of the diffraction edges – Fraunhofer diffraction by a single slit - Fraunhofer diffraction by multi slits – diffraction grating – Resolving and dispersion power – Polarization – Polarization by reflection and refraction – Polarization by double refraction – Nicol prism – Hygens theory for Polarization of light by double refraction - Polaremeter.

References:

- Available text books at the library and digital library.
- Soars, John and Liz Soars, Headway: Elementary level Oxford University 1996. □

PHYS212 Advanced Optics:

2 Credit hrs. Lecture + 1 Credit hr. Practical

Prerequisites Courses: PHYS211

Maxwell's equations - Maxwell's equations and the properties of light wave - The equation of wave motion - Energy in a wave and Poynting theorem - Reflection and

refraction of electromagnetic wave - Fresnel's equation - Brewster's law - The polarization - Polarization by reflection - Polarization by refraction - Double refraction and explain Huygens's theory-Different method of polarization - Elliptically and circularly polarized light for uniaxial crystal - wave surface for ordinary beam and extraordinary beam - Babinet's compensator - The quarter wave plate - Faraday effect - Kerr effect-Dispersion - Normal dispersion and Cauchy's equation - Anomalous dispersion and Sellmeier's equation - Optical activity.

References:

- Available text books at the library and digital library.
- Soars, John and Liz Soars, Headway: Elementary level Oxford University 1996.

PHYS213 Vibration and Waves:

2 Credit hrs. Lecture

Prerequisites Courses: PHYS102

Simple harmonic motion – Free vibrations – Oscillatory – Resonance – Superposition of harmonic vibrations – One dimensional treatment of waves – Differential form of the wave equation – Velocity of waves in strings, rods and gases – Group velocity – Standing waves – Elementary consideration of diffraction of x-rays and electrons by crystalline materials – Waves in more than one dimension – Water waves – Solitary waves.

PHYS221 Astrophysics and Meteorology Physics:

2 Credit hrs. Lecture

Prerequisites Courses: PHYS101

Light- stellar distances - Analyzing starlight - Stellar properties (luminosity and masses- photometric terms - Forms of radiation- temperature degrees) - Stellar spectra - Galaxies - Basic properties of the planetary system - Planetary atmospheres - Modeling the atmospheres of planet.

PHYS222 Thermodynamics:

2 Credit hrs. Lecture + 1 Credit hr. Practical

Prerequisites Courses: PHYS101

Basis of thermodynamics- kinetic theory of gases- equations of states for ideal gases- vander Waals equation for real gas - Boyle Degree of Real Gas - Andraw's equation- state variables- Work- internal energy- Heat capacity- Differentiation of state function-

Volume expansion coefficient and compressibility- The first law of thermodynamics- Applications of the first law of thermodynamics- The second law of thermodynamics- Applications of the second law of thermodynamics- Entropy- Maxwell- Boltzmann's equation distribution- Second equation of energy- Joule- Thomson experiment- Joule effect- Reversible and irreversible processes the first law of thermodynamics- thermodynamics potentials- Phase change- Third law of thermodynamics- Applications on the third law of thermodynamics- Heat transfer in solids- Black- body radiation and Steeven's law- Distribution function- Distribution of molecule's velocity- distribution functions of energy- Probability functions.

References:

- Heat and Thermodynamics , Zemansky&Dittman, McGrew-Hill.
- Physics:Principle with applications d.c Gancoli,U.S.A.

PHYS223 Crystallography and x- ray Diffraction:

2 Credit hrs. Lecture

Prerequisites Courses: PHYS231

Types of solids - Types of atomic bonds - Methods of crystal growth (Predgeman method - Zohker Alexy method- Partial melting method) - Crystal structure- Bravais lattice - Miller indices - Crystal imperfections - Bragg's law - Diffraction methods (Laue method- Powder method - Rotation method) - Electrons diffraction - Neutrons diffraction - Lattice dynamics and photons - Lattice vibrations - Thermal properties of dielectric materials.

PHYS231 Atomic Physics:

2 Credit hrs. Lecture + 1 Credit hr. Practical

Prerequisites Courses: PHYS101

Bohr's theory of the hydrogen atom – Motion of the hydrogen nucleus – Elliptic orbits for hydrogen – Optical spectral series – Vector model of atom – Pauli's exclusion principle – Spectral notation – Spectrum of sodium – Zeem effect (normal - anomalous) – Land'e g factor – Stern – Gerlach – Spectra of two electron atoms – the laser – X-ray.

References:

- Available text books at the library and digital library.

PHYS241 Electromagnetic:

2 Credit hrs. Lecture + 1 Credit hr. Practical

Prerequisites Courses: PHYS102

Waves - Calculated the differential and integrating waves - Semi axis - electrostatic - Electrostatic field and its properties - Coulomb Force - Intensity of the electric field -

Electrical potential - Work and the energy in the electrostatic field - Laplace's Equation - solution of Laplace's equation - Electric field equation of forces lines - Equation of equal potential surfaces - Electrical polarizability and its applications - Applications of the dipole and its moment - Quarter dipole - Electrostatic field in the materials - The electric current and the conductors - Ohm's law - Continuity equation - Magnetic field - Ampere's law - Faraday's law and the induction force - Lorentz equation - The motion of conductor in magnetic field - Maxwell equation - Hall effect - The magnetic properties of Paramagnetic and diamagnetic materials - Magnetic susceptibility - Hysteresis loop - Magnetic surface.

References:

- Engineering electromagnetics, McGraw-Hill, William H. Hayt.
- Fundamentals of physics by David Halliday and Robert Resnick.

PHYS242 Mathematical Physics (1):

2 Credit hrs. Lecture

Prerequisites Courses: PHYS102

Introduction – Discovery of Errors – Error Correction – Solution of linear Algebraic Equation – Inverse Matrices – Vectors – Multiplication of Vectors. Gradation and Division – Gauss Jordan Method of Solving Linear equation – Inverse Matrix using Gauss Jordan Method - Linear Integration – Stokes – Numerical Integration - Fourier Series and Applications – Differentiation and Integration of Fourier Series – Solving a number of problems using Fourier Series.

References:

- Computer methods for science and engineering. Robert L. Lawless, International text book company (1973).
- Mathematical methods for physicists, George Arfken (1970).

PHYS243 Modern Physics:

2 Credit hrs. Lecture

Prerequisites Courses: PHYS241

Particle nature of waves: The Photo electric effect - Quantum theory - Wave Particle - Duality - X-ray - X-ray diffraction - The Compton effect - Pair Production. Wave nature of particles: De Broglie Wave - Wave Function - De Broglie Wave Velocity - Phase and group velocities - Electron diffraction - Uncertainty Principle - Application of the Uncertainty Principle - Atomic Structure: Atomic models - Rutherford Scattering

Formula - Electron Orbits - Atomic Spectra - Bohr Theory for the atomic structure - Energy levels and spectra - Atomic Excitation - The Quantum numbers.

References:

- Concept of modern physics (Arthur Beiser).
- Modern physics for scientists and engineers (Stephen T. Thornton).

PHYS251 Principals of Electronic Circuits :

2 Credit hrs. Lecture + 1 Credit hr. Practical

Prerequisites Courses: PHYS102

D.C. current circuits: series and parallel circuits - Kirchhoff's laws - Circuit analysis systems (general method- node's analysis - equivalent voltage sources conversion for node analysis - the superposition theorem - Thevenin's theorem - Norton's theorem) - A.C. circuits: waveform shapes - Effective value - Voltage and current in R - L - C - Power and power factor correction - Complex numbers - complex reactance -Conduction in series and parallel in A.C. circuits - Resonance circuits-Four-Terminal Network: - Z-parameters - H-parameters - Y-parameters - Equivalent circuits- Study of integrator and differentiator - Low pass and high pass filters - Frequency response - Oscilloscope.

References:

- "Applied electricity for engineers" By: Boris Kuznetsov, Mir Publishers-Moscow.

PHYS311 Atomic Spectroscopy:

2 Credit hrs. Lecture + 1 Credit hr. Practical

Prerequisites Courses: PHYS231

Emission and absorption spectra - Concept for spectral line intensity - Broadening of spectral line - Factor affecting the broadening of spectral line - Dispersion and resolving power - Dispersion power and deviation for prism and diffraction grating - Different type of spectrum devices - Excitation sources - The quantitative and qualitative analysis - Standard calibration curve and working curve.

References:

- An introduction to atomic and nuclear physics, H. Semat

PHYS312 Electron optics:

2 Credit hrs. Lecture

Prerequisites Courses: PHYS211

Introduction to electron optics – Motion of electrons in uniform (electrostatic and

magnetic) fields – Types and design of electron lenses – Physical similarity of light and electron lenses – Types of electron microscope and applications.

References:

- Principle of electron optics Hawkes PW, Kasper (1996).

PHYS321 Solid State Physics (1):

2 Credit hrs. Lecture + 1 Credit hr. Practical

Prerequisites Courses: PHYS231

Basis of crystal structure - One dimensional lattice - Two dimensional lattice - Three dimensional crystal lattice - Primitive cell and unit cell - Symmetry operations - Crystal lattice systems - Bravais lattice - Miller indices - Examples of the crystal structures - Identification of the crystal structure by x-ray diffraction - Bragg's law - Electrons diffraction - Neutrons diffraction - Diffraction methods for crystal structure analysis (Laue method - Powder method - Rotation method) - Crystal binding - Crystals of inert gases - Van der Waals interaction - Repulsive energy - Attraction energy - Ionic crystals and Madelung constant - Covalent crystals - Metal crystals - Hydrogen bonded crystals - Thermal properties of solids - Heat capacity - Classical theory - Einstein theory - Debye theory - Heat capacity of conduction electrons in metals – Phonons - Thermal conductivity of solids - Thermal expansion - free electron theory of metals - Drude and Lorentz classical theory - Sommerfeld quantum theory - Electrical conductivity and Ohm's law Wiedemann and Franz ratio - Energy levels and density of orbital in one dimension - density of orbital in three dimension and Fermi energy level- Fermi - Dirac distribution function - heat capacity of electron gas - Fermi- Dirac distribution effect on the electrical conductivity - Thermo ionic emission - Photoelectric effect of metals - Crystal defects- Point imperfection (impurities- Schottky defect and Frenkel defect) - Diffusion- color centers- Line imperfections - dislocations.

Reference:

- Elementary solid state physics ,Omar

- Solid state physics , A.J. Dekker

PHYS322 Ultrasonic Waves:

2 Credit hrs. Lecture

Prerequisites Courses: PHYS213

Fundamentals of ultrasonic waves - Equation of sound wave - Acoustic parameters - acoustic impedance - Acoustic intensity-acoustic power - Reflection and transmission of

ultrasonic waves on boundary at normal incidence and oblique incidence - Mode conversion attenuation of ultrasonic waves - Ultrasonic transducers - Piezoelectric transducer - Magnetostriction transducer - Fringes of ultrasonic waves - Ultrasonic cavitation - Factors affecting ultrasonic cavitation intensity - High intensity ultrasonic applications (cleaning - Electroplating - Aerosol production - Welding metals - Chemical reactions - Emulsification - Homogeneous distribution) - Low intensity ultrasonic applications (elastic parameters - velocity of a moving fluid by different methods - viscosity measurements).

Reference:

- "Ultrasonic", by Benson Carlin.

PHYS323 Environmental Physics:

2 Credit hrs. Lecture

Prerequisites Courses: PHYS101

Heat transfer - Thermal radiation - Solar energy and its technology - Molecular absorption of electromagnetic waves - The earth's thermal environment - Green house phenomenon - Effect of nuclear explosions - Water vapor and humidity - Environmental pollution - Ozon layer - Air pollutant - Noise and light pollution- Effect of different radiation on humans - Radiation protection.

References:

- Principle of Environmental physics by,Moneteith,J.L and Unsworth M.

PHYS324 Solid State Physics (2):

2 Credit hrs. Lecture + 1 Credit hr. Practical

Prerequisites Courses: PHYS321

Band theory of solids- Bloch theorem- Kronig Penny model - Effective mass - Velocity and acceleration of electrons in solids - Brillouin zones - Classification of solids: conductors - Semiconductors and insulators - Intrinsic and extrinsic semiconductors - energy bands of electrons for intrinsic and extrinsic semiconductors - Fermi energy level - Carriers concentration in intrinsic and extrinsic semiconductors - Effect of temperature on semiconductor resistivity - Mobility and electrical conduction of intrinsic and extrinsic semiconductors motion of electrons in a magnetic field and Hall effect - Contact phenomenon: contact between two different metals - Metal -semiconductor contact - P-n semiconductors junction - Rectification properties of p-n junction and its applications - Zener diode - Photodiode - Tunnel diode and transistors.

References:

- Elementary solid state physics ,Omar.
- Solid state physics , A.J. Dekker.
- Introduction to solid state physics , Charles Kittel.

PHYS325 Solid State Physics:

3 Credit hrs. Lecture + 1 Credit hr. Practical

Prerequisites Courses: PHYS231

Basis of crystal structure - One dimensional lattice - Two dimensional lattice - Three dimensional crystal lattice - Primitive cell and unit cell - Symmetry operations - crystal lattice systems - Bravais lattice - Miller indices - Examples of the crystal structures - Identification of the crystal structure by x- ray diffraction - Bragg's law- electrons diffraction - Neutrons diffraction - Diffraction methods for crystal structure analysis (Laue method - powder method - rotation method) - Crystal binding - Crystals of inert gases - Van der Waals interaction - repulsive energy- Attraction energy - Ionic - Metallic - Energy and Modelung constant- Covalent crystals - Metal crystals- Hydrogen energy bonded crystals - Thermal properties of solids - Heat capacity - Classical theory - Einstein theory - Cebye theory - Heat capacity of conduction electrons in metals - Phonons - thermal conductivity of solids - Thermal expansion - Free electron theory of metals - Drude and Lorentz classical theory - Summerfield quantum theory - Electrical conductivity and Ohm's law Wiedmann and Franz ratio - Energy levels and density of orbital in one dimension - Density of orbital in three dimension and Fermi energy level - Fermi - Dirac distribution function - Heat capacity of electron gas - Fermi - Dirac distribution effect on the electrical - Band theory of solids- Bloch theorem- Kronig Penny model- effective mass - Velocity and acceleration of electrons in solids - Brillouin zones- Classification of solids: conductors - Semiconductors and insulators - Intrinsic and extrinsic semiconductors - Energy bands of electrons for intrinsic and extrinsic semiconductors - Fermi energy level- Carriers concentration in intrinsic and extrinsic semiconductors- Effect of temperature on semiconductor resistivity - Mobility and electrical conduction of intrinsic and extrinsic semiconductors motion of electrons in a magnetic field and Hall effect - Contact phenomenon: contact between two different metals - Metal-semiconductor contact - P-n semiconductors junction - Rectification properties of p-n junction and its applications - Zener diode - photodiode - Tunnel diode and transistors.

References:

- Elementary solid state physics ,Omar.
- Solid state physics , A.J. Dekker.
- Introduction to solid state physics , Charles Kittel.

PHYS331 Nuclear Physics (1):

2 Credit hrs. Lecture + 1 Credit hr. Practical

Prerequisites Courses: PHYS231

Principal nuclear properties – Binding energy – Natural and artificial nuclear disintegration – Production of nuclear isotopes by nuclear bombardment – Chain disintegration – Different types of equilibrium – Radioactive decay law – Radioactive decays (γ decay – internal conversion - α decay - β decay)- Neutrino hypothesis – electron capture – Interaction of γ rays with matter – Nuclear detectors.

References:

- Nuclear physics by I. Kaplan

PHYS332 Accelerators:

2 Credit hrs. Lecture + 1 Credit hr. Practical

Prerequisites Courses: PHYS331

The Cockcroft – Walton machine – The electrostatic generator or Van de Graaff machine – The cyclotron – The frequency-Modulated cyclotron or synchrocyclotron – The acceleration of electrons - The betatron and electron synchrotron – The proton synchrotron – Linear accelerators – The alternating-gradient synchrotron – Colliding accelerator.

References:

- Nuclear physics and nuclear reactor by A.Klimov.
- Nuclear physics, an introduction by SB Patel.

PHYS333 Plasma Physics:

2 Credit hrs. Lecture

Prerequisites Courses: PHYS231

Definition of plasma – Plasma temperature and particle energy – Motion of a charged particle in an electromagnetic field – Particle drift in a (diverging – converging oscillating) magnetic field.

References:

- Introduction to plasma physics and controlled fusion by F.Chen.
- Introduction to plasma theory by D.R. Nichlson.

PHYS334 Gamma spectroscopy:

2 Credit hrs. Lecture

Prerequisites Courses: PHYS331

Decay schemes for radioactive nuclei – Selection rules for γ transitions – Scintillation detectors – Semi conductor detectors – Components of a single γ - ray spectrometer - Calibration and resolution – Types of spectra.

References:

- $\alpha\beta\gamma$ spectroscopy by K.Sigban.
- T.Sovlfanidis N., "Measurements and detection of radiation", 1983.

PHYS341 Quantum Mechanics (1):

2 Credit hrs. Lecture

Prerequisites Courses: PHYS241

A historical introduction (Theories and Experiments) – Modern Quantum Theory – Assumption in Quantum Theory – Operators – Hermethien Operators – Schrodinger Equations in one and three dimensions – Properties of Schrodinger's Equation – Orbital Angular Momentum and interrelated relations – Applications on Quantum Mechanics (Infinite Well) – (Quantum Well). Scattering of particles from Potential Barrier - Scattering of particles from finite potential barrier.

References:

- Quantum Mechanics by Hisberg and Remick.
- Modern physics and quantum mechnics by Anderson.

PHYS342 Statistical Physics :

2 Credit hrs. Lecture + 1 Credit hr. Practical

Prerequisites Courses: PHYS242

Basic and supplementary energy states – Micro and Macro states – Thermodynamic probabilities (Maxwell - Boltzman Probabilities – Fermi- Dirac Probabilities – Bose – Einstein Probabilities) – Fermi-Dirac Statistics – Statistics of Entropy – Partition Function - Using Partition Function to solve general Constants of Materials – Applications of Material in its different states.

References:

- Statistical physics by F.Reif.
- Thermodynamics,and Statistical mechanics by Addison-wesly (1972).

PHYS343 Electrodynamics:

2 Credit hrs. Lecture

Prerequisites Courses: PHYS241

Maxwell Equation of Electromagnetic waves – Maxwell Equation of Free Space – Solution of Maxwell Equation using Potential – Gig Conversion – Lawrence Condition and Potential Equation- Law of Energy Conservation in Electromagnetic field (Poynting Theory) – General solution of Maxwell Equation in space – General Equation of Waves – Electromagnetic waves in dielectric media – Polarization - Conditions of neighboring media – Reflection and Refraction Law in Electromagnetic field - Electromagnetic in Conducting Media – Guided and hollow waves - Relativistic form of electrodynamics.

References:

- Introduction to electrodynamics by David J.

PHYS351 Electronics (1):

2 Credit hrs. Lecture + 1 Credit hr. Practical

Prerequisites Courses: PHYS251

A.C. circuits: Waveform shapes - Effective value - Voltage and current in R - L - C- power and power factor correction - Complex numbers - Conduction in series and parallel in A.C. circuits - Resonance circuits - Four-Terminal Network: - Z-parameters - h-parameters - y-parameters - Equivalent circuits: integrator and differentiator circuits - Low pass and high pass filters - Frequency response - Oscilloscope. Theory of energy levels: n-type semiconductor and p-type semiconductor - Fermi level - Acceptor and donor energy levels - Current carriers and life time - Electric conduction in semiconductors.

References:

- "Integrated electronics: analog and digital circuits and systems" By: Jacob Millman, McGraw-Hill Inc. Japan.
- "Applied electricity for engineers" By: Boris Kuznetsov, Mir Publishers-Moscow.

PHYS352 Electronics (2):

2 Credit hrs. Lecture + 1 Credit hr. Practical

Prerequisites Courses: PHYS351

P-N junction diode - Forward and reverse bias - Characteristic curve - Zener diode - tunnel diode - Photo diode and photo emission- Half and full wave rectification - Filters -

BJT as four terminal network for small signals - Equivalent circuit of bipolar transistor - Some application of BJT transistor - Transistor as a current and voltage amplifiers - Field effect transistor: operation principle- Transistor parameters - Characteristic curves - Feedback- Oscillators - Operational amplifier - Some of their characteristics and applications.

References:

- "Integrated electronics: analog and digital circuits and systems" By: Jacob Millman, McGraw-Hill Inc. Japan.

PHYS353 Electronics:

2 Credit hrs. Lecture + 1 Credit hr. Practical

Prerequisites Courses: PHYS251

P-N junction diode - Forward and reverse bias - Characteristic curve - Zener diode- Tunnel diode - Photo diode and photo emission - Half and full wave rectification - Filters - BJT as four terminal network for small signals - Equivalent circuit of bipolar transistor - Some application of BJT transistor - Transistor as a current and voltage amplifiers - Transistor parameters - Characteristic curves – Feedback – Oscillators - Operational amplifier - Some of their characteristics and applications.

References:

- "Integrated electronics: analog and digital circuits and systems" By: Jacob Millman, McGraw-Hill Inc. Japan.

PHYS411 Laser's physics and its applications:

2 Credit hrs. Lecture + 1 Credit hr. Practical

Prerequisites Courses: PHYS311

Simulated and spontaneous emissions – Einstein's equations - Amplification in an active medium and growth of a laser beam – Optical resonator theory – Methods of producing inverted population – Energy level diagram for lasing action – Two, three, four level systems – types of lasers (gas lasers, semiconductor lasers, molecular laser, high power lasers) – Properties of laser beam – Losses and its types - Laser gain – Single mode operation – Optical feedback – Laser applications such as (industrial applications, medical applications, military applications, daily applications, scientific research applications, special applications).

References:

- Laser fundamentals by W.T. Silfvast, Cambridge

PHYS412 Molecular Spectroscopy:

2 Credit hrs. Lecture

Prerequisites Courses: PHYS311

Electromagnetic radiation – The interaction of electromagnetic radiation with atom & molecules – Microwave rotational spectra – Rotational molecules – Rotational spectra of diatomic molecules – Polyatomic molecules at its spectra – Techniques and instrumentation – Infra red spectroscopy – Vibrating diatomic molecules – Vibrational – Rotational spectra – Techniques and instrumentation.

References:

- "Fundamentals of Molecular Spectroscopy", C.N Banwell, MC Graw-Hill Book Company (uk) Limited.
- "Introduction to Molecular Spectroscopy", G.M. Barrow, MC Graw-Hill international Book Company.

PHYS421 Materials Science (1):

2 Credit hrs. Lecture + 1 Credit hr. Practical

Prerequisites Courses: PHYS324 or PHYS325

Mechanical properties: the stress- the strain - Types of stress- Hock's law and elasticity factors (linear and volumetric elasticity) - Tension tests- Stress strain curve for metals - Stress strain relation for low carbon materials - Creep tests- Factors affecting creep and fatigue. Optical properties: light as an electromagnetic waves and photons - Light intensity - refractive Index - Dielectric constant - Optical conductivity - Absorption of light and absorption coefficient- Reflectivity- Transmission- Calculation of the optical constants - Transition in fundamental absorption edge- Permissible and non permissible transition - Direct and indirect transition- Absorption by excetons - Absorption by impurities - Absorption by free carriers. Thermal properties: Lattice motion – phonons - thermal conductivity by different methods - Heat transfer in insulators - Thermal properties in solids - Heat transfer by different methods- Instantaneous thermal conduction - Thermal diffusion coefficient - Thermal expansion in solids.

References:

- "Material science and engineering", an introduction by WilliamD, Callisters.

PHYS422 Magnetic Resonance:

2 Credit hrs. Lecture

Prerequisites Courses: PHYS324 or PHYS325

The resonance phenomenon - Electron's spin and spin magnetic moment - Energy of free electron in a magnetic field - Electron paramagnetic resonance spectrum - Hamiltonian and wave function - The complete Hamiltonian operator - The Zeeman hyperfine spectrum - Spin-spin interaction in a crystal field - Spin - orbital angular coupling - Nuclear Zeeman effect - Theory of relation and line shapes - Electron spin echo - Neutron magnetic resonance NMR spectrum - Origin of spectrum - The chemical shift- The physical mechanism underlying chemical shift line multiplicity in NMR- Analysis of spin systems - The line-widths of the nuclear magnetic resonance spectra - Magnetic noise - The Bloch formulations.

References:

- "Electrons spin resonance theory & applications", by N.M.A. theraton (1973).

PHYS423 Solar Energy:

2 Credit hrs. Lecture

Prerequisites Courses: PHYS321 or PHYS325

Energy sources - sun as an energy source - Sunlight - Semiconductor principles - Optical and photoelectrical properties of semiconductors – Photo static effect - Low resistivity materials- Dember photo effect – Photoconductivity - Photovoltaic effect in p-n junction and p-n hetero junction (junction capacity and depletion width - characteristics - V_{oc} - J_{sc} - efficiency) - Metal semiconductor junction - Conversion of solar energy into electricity - Photovoltaic effect - The solar cell and it's fabrication - Examples of p-n junction and p-N hetero junction solar cells - Photosensitive – Photo emissive – Photo galvanic – photo magnetic) - Indirect energy conversion into electricity (Through a heat stage - through a chemical stage - Photosynthesis).

References:

- "Solar energy conversion", by Richard C.

PHYS424 Nanotechnology:

2 Credit hrs. Lecture

Prerequisites Courses: PHYS321 or PHYS325

A brief history of nano revolution - Classes of nanostructure materials - Synthetic strategies- Size control - Shape control - Physical properties of nano materials - Chemical

properties of nano sited materials - Methods of preparation of nano materials- Chemical methods- Physical methods - Nano materials characterization - X-ray characterization of nano particles - Scanning electron microscope - Field emission electron microscope - transmission electron microscope - Atomic force microscope - Scanning tunneling microscope - Fourier transformation infrared spectrometry - Raman spectroscopy - X-ray fluorescence (XRF) - X- ray photoelectron spectroscopy (XPS) - Energy dispersive x- ray spectrometry (EDS).

PHYS425 Physics of Amorphous Materials:

2 Credit hrs. Lecture

Prerequisites Courses: PHYS324 or PHYS325

Polymers and plastics - Methods of preparation of polymers - Examples of polymers - Ceramics - Dielectric constants of different polymers - The properties of ceramics – Piezoelectricity – Ferro electricity - Ferromagnetism ferrites and superconductors - Liquid crystals- Electrolytes - Fiber optics - Fiber glasses - Silicon and amorphous semiconductors - Synthesized materials.

References:

- "Amorphous semiconductor".

PHYS426 Materials Science (2):

2 Credit hrs. Lecture + 1 Credit hr. Practical

Prerequisites Courses: PHYS421

Electrical properties of the materials - Equilibrium state of the electronic gas in conducting materials in absence of the electric field - Deviation of the electron in presence of the electric field - Relaxation time and mean Free path - Specific conductance in conductors - Electrical conductivity in ideal gas and non ideal gas - Widman Lorentz law - Dependence of free carriers on temperature - Electrical conductivity in pure metal – Alloys and extrinsic semiconductor - Magnetic properties of materials - Origin of magnetic phenomena - Classification of magnetic materials - Magnetic moment of electrons - Classical paramagnetic theory - Paramagnetic materials - Magnetization curve - Magnetic domain - Exchange energy – anti Ferro magnetism - Super conductivity of material - Discovery of Superconductivity - Critical magnetic field - Zener effect - Heat capacity - Energy gap for supper conductors and its measurements - Isotopic effect - London equation - Bandon and Coops theory - Second type of

superconductor current - Quantum magnetic fluctuation - High temperature superconductor - Mechanism of the High temperature superconductivity – Applications.

References:

Material science and engineering, an introduction by William D. Callister

PHYS431 Nuclear Physics (2):

2 Credit hrs. Lecture + 1 Credit hr. Practical

Prerequisites Courses: PHYS331

Nuclear reactions and theories – effective cross section – nuclear forces – scattering in center of mass & laboratory coordinates and the relation between them in an elastic scattering – nuclear models (shell model – liquid drop model – collective model).

References:

- Fujia Yang and Joseph H. Hamittan, "Modern Atomic and Nuclear Physics".1996.
- I.S Hughes, "Elementary Particles", third edition 1991.
- $\alpha\beta\gamma$ spectroscopy by K. Sigban.

PHYS432 Radiation Physics (1):

2 Credit hrs. Lecture

Prerequisites Courses: PHYS331

Radiation sources: (fast electrons – Heavy charged particles – Electromagnetic radiation – neutrons) – Radiation units - Interaction of radiation with matter (interaction of heavy charged particles – fast electrons and neutrons) – Radiation doses – External and internal hazards of radiation - Effective factors(time - distance – shielding) – Protection from ionizing radiation and dealing with radioactive exposures such as to radioactive iodine and contamination of skin with radioactive substances – applications in medicine - Agriculture and industry.

References:

- "Elementary introduction to nuclear reactor" by S.E. Liver

PHYS433 Nuclear Reactors and Safety:

2 Credit hrs. Lecture + 1 Credit hr. Practical

Prerequisites Courses: PHYS431

Neutrons – Neutron reactions – Fission – Fission produced neutrons and cross section – Effect of neutron velocity on fission reaction – Neutron cycle – conservation of energy – Nuclear fission reactors – Components of a fission reactor (fuel – moderator – cooler – control rods – safety rods –neutron reflector) – Different types of reactors for electricity

generation (boiling water reactors – compressed water reactors – heavy water reactors – gaseous cooling reactors – graphite and water reactors) – Research reactors – Breeding reactors – Neutron flux calculation – Calculating the critical size for a reactor - Logarithmic rate for the moderation of energy – Moderation power –Radiation dosimeters – Biological effects due to neutrons.

References:

- "Nuclear physics and nuclear reactors" by V.L.Lefi.

PHYS434 Radiation Physics (2):

2 Credit hrs. Lecture

Prerequisites Courses: PHYS432

Biological effects of radiation – Interaction of radiation with cells – Effect of the radiation on the body and genes – Cosmic rays – Earth radiation – Radioactive sources used in medicine for diagnosis and therapy – Sources use in industry and wars – Radiation protection in medicine – Basic fundamentals for protection from opened and closed sources – Waste products: liquid - solid - and gases waste products and how to deal with them.

References:

- "Elementary introduction to nuclear reactor", by S.E. Liver.

PHYS441 Quantum Mechanics (2):

2 Credit hrs. Lecture

Prerequisites Courses: PHYS341

Matrix Theory of Harmonic Oscillator – Central Potential and Hydrogen Atoms – Quantum Numbers – Angular Momentum of Electronics – Motion and Periodic table – Perturbation Theory independent of time – Application on Helium Atom – Perturbation dependent on time – Relativistic Schrodinger Equation – Klien-Gordon Equation.

References:

- Quantum mechanics (Schiff).
- Quantum mechanics by Greiner.

PHYS442 Elementary particle Physics:

2 Credit hrs. Lecture

Prerequisites Courses: PHYS331

Particles and anti-particles- Leptons- Quarks-Gluons- Hadrons- Electro-magnetic force- weak nuclear force- electroweak force- strong nuclear force- Grand Unified Theory

(GUT) – Gravitational and Quantum Gravity – Super force – the Big Bang Theory- Conservation Laws (Lepton number-baryon number- charge conjugation – Strangeness) – Classification of Particles by: (interaction- decay product – speed of decay – spin – Isospin – Parity – charge conjugation – time reversal) – Quark model of baryons and mesons- The eightfold way – Four vectors – Strange particles.

References:

- An introduction to quarks and partons by F. E.Close.
- Elementary particles by I.S.Hughes.

PHYS443 Mathematical Physics (2):

2 Credit hrs. Lecture

Prerequisites Courses: PHYS242

Numerical Solution of differential Equations – Euler's Method – Modified Euler's Method – Picard's Method – Numerical Integration – Trapezoidal Rule – Taylor's Series Method – Runge –Kutta Method – Runge – Kutta Method for Second – order Differential Equation – Solving differential equations by Inverse operators – Sturm – Liouville Theory of Differential equations Finite difference Methods – Taylor's Series (Numerical differentiation and integration) – Linear difference equations with constant coefficient – Laplace Transform.

References:

- Computer methods for science and engineering. Robert L.Lafara, International text bookcomp.lim.(1973).
- Mathematical methods for physicists, George Arfken (1970).

PHYS451 Logic Circuits:

2 Credit hrs. Lecture + 1 Credit hr. Practical

Prerequisites Courses: PHYS352 or PHYS353

Digital techniques - number systems- logic gates - Boolean algebra and logic theorems - Karnaugh maps - Integrated circuits - The basic logic gates circuits by using bipolar and unipolar transistors - Flip-Flops - Counters - 7-segment circuit and decoder driver circuit - Registers.

References:

- "Integrated electronics: analog and digital circuits and systems" By: Jacob Millman, Mc Graw-Hill Inc. Japan.

PHYS171 General Biophysics (1):

2 Credit hrs. Lecture + 1 Credit hr. Practical

Prerequisites Courses: -

During the bio-Electric body-nervous system and Neuron- Voltage nerve- Heart rate- Making the brain-making the eye. X-ray diagnostic physics: the generation of X-ray and absorbed- X-rays. Radioactive sources used in nuclear medicine- The statistical aspects of nuclear medicine. Radioactive core instrumentation and medical applications: as Jaeger- tube replication photogravure- analyst with the high pulse- gamma camera- radiation doses in nuclear medicine. Prevention of radiation in medicine: natural and industrial sources of ionizing radiation- biological effects of ionizing radiation- units of radiation protection and nuclear medicine in its heat and cold applications in medicine: warming treatment using both- the way we communicate gases- infrared and radio waves, waves and waves over Almkronip Sound- the use of extremely low temperatures in medicine- keeping blood and tissue surgery and refrigeration. The use of light in medicine: characteristics of the Adwae and measurement units light- visible light applications in medicine- perspectives flexible optical fiber applications- ultraviolet and infrared in medicine- infrared imaging- Almicroscobate.

References:

- "Fundamentals of Ultrasonographic Techniques" By: J.D Wicks & K.S. Howe.
- "Aspects of Biophysics" By: William Hughes.

PHYS172 General Biophysics (2):

2 Credit hrs. Lecture + 1 Credit hr. Practical

Prerequisites Courses: PHYS171

The main energy and power within the body- Keeping energy in the body- Replacing the body energy- Labor and power systems essential to have thermal energy body. Physics eye and vision- Heralds a sense of vision and visual properties of private collection agents photo eye. Network member Adwae sensitive to eye- Species overall light receptors. And correction of vision defects- Eye lenses and efficient vision. Sound in medicine- General characteristics of voice- Reflection of sound waves- Alstioskub (stethoscope)- The body imaging ultrasound waves- Doppler effects- The physics of the ear and hear in- Hearing test- Deafness and audio assistance- Biophysics muscle Structural flexibility- The impact of loads on tonicity- Muscle tension- Dynamic blood. Hydraulic principle of force and its relationship to the filling of the heart. Heartfelt flow and blood pressure. landkest phenomenon. Bozail equation Bernoulli equation in relation

to high blood pressure. Vital lung mechanics- Surface tension and lung suspended. The impact of detergent on the process of breathing. The relationship between force and air pressure and resistance during the process of breathing.

References:

- "Basic Physics and Technology of Medical Diagnostic Ultrasound" By: M.Husse.
- "Medical Physics" By: O. Glasser, J.R. Cameron & J.Shofronick.
- "Biomechanics Its Foundation & Objectives" By: Y.C.Fung , N.Perrone & M.Anliker.

PHYS271 Environmental and general Biophysics:

2 Credit hrs. Lecture + 1 Credit hr. Practical

Prerequisites Courses: PHYS172

Study and the various functions of the body- The transfer of some ions and molecules in the cell membrane- Electromagnetic waves Living Catalog studies on the living cell- The flexibility of a living cell- The wife of live material of the cell- The cell surface energies- Biological effects of some areas- The transition and absorption- The physical foundations of the highlights of vital effects Vital clinical curative effects- Electromagnetic waves(non ionizing) spectrum of areas- Measuring vital resistance- The effects of low frequency- microwave radiation- The definition of environmental pollution- The types of environmental contaminants- Environmental pollution by radiation- Radioactive sources (solid and liquid) natural radioactive materials.

References:

- "Dielectric Behavior of Biological Molecules in Solution" By: Grant, R.J. Sheppard and G.P. South.
- "Concepts of Radiation Dosimetry" By: Kase & Nelson.

PHYS272 Biophysics Waves:

2 Credit hrs. Lecture + 1 Credit hr. Practical

Prerequisites Courses: PHYS271

Wave mechanics- Physical properties- The properties of the center-Proliferation- The spread of the models sound like- Sound waves interact with biological tissue (Reflection - scattering - Refraction - Diffraction - Interference - Absorption)- The sound intensity in the tissue- The appointment of the severity of the sound echoed in the tissues- Applications of ultrasound in medical imaging using ultrasound. light waves: The nature and sources of light - Perspectives and X-ray and uses in medicine- And the deviation

Polarization photogravure Appliances- Optical bio-medicine- Overlapping and coral and Refractive experience overlap Yang- Haegn theory- Quantum theory of the Bohr atom of hydrogen- Light dispersion and break it and use them to learn volumes Biological macromolecules- Activity optical phenomena are connotation of biophysical sound units used in lasers- UV radiation measurements of a voice- Acoustic generators- considerations of the physical audio-ray absorption.

References:

- "Introduction to Classical and Modern Optics" By: Meyer Arenot.
- "Laser in Biology and Medicine" By: F. Hillenkamp, R. Partesi C.A. Sacchi.

PHYS273 Bio-Potential:

2 Credit hrs. Lecture

Prerequisites Courses: -

Flood of energy in the world of biological change in the free energy of chemical reactions under conditions Standard- of a system. A. T. B. And the transfer of the cell- Almitokondria, of a given. A. T. B cell- and breathing through oxidative Alphusfarh- Alblastaidat- construction and optical Alphusfarh optic- an enzyme- a construction to A. T. B. - The principle of osmosis circles chemical- protein, employment is vital to build chemical- construction h vital. D. N. And proteins- the cellular assembly installation.

References:

- "Aspects of Biophysics" By: William Hughes.

PHYS274 Thermodynamic of Biological Systems:

2 Credit hrs. Lecture + 1 Credit hr. Practical

Prerequisites Courses: PHYS101

The first law of thermodynamics and its application to biological systems- The internal energy to work- First law, the inthalpy- the ideal situation- Applications to large molecules- Heat capacity- The survival of energy living organisms. Second law of thermodynamic: The entropy- Thermal energy- The same heat- Breaking protein- Third law and biology- Transitions and life. Gibbs free energy- Balanced situation- Shifting operations- Transition phase- The chemical potential- The ionic liquids- The ideal situation for large molecules- The impact of temperature on stable equilibrium- Chemical association of biological molecules. Application Gibbs free energy: Photosynthesis- osmosis pressure- Dunan balance- To move through the membrane- The interaction of enzymes- And aid dissolving the proteins- Their balance and their dynamics.

References:

- "Dielectric Behavior of Biological Molecules in Solution" By: Grant, R.J. Sheppard and G.P. South.

PHYS371 Bio statistics:

2 Credit hrs. Lecture + 1 Credit hr. Practical

Prerequisites Courses: -

Concept of bio statistics – Central limit theorem – Methods of deviation, variance and standard deviation – Elementary probability – Probability distribution and types of statistical population – Hypothesis testing – Linear regression and correlation – Health bio statistic.

References:

- "Mathematical Methods in Physics Science" BY, Boas.

PHYS372 Electronics and electricity of Bio Systems:

2 Credit hrs. Lecture + 1 Credit hr. Practical

Prerequisites Courses: PHYS251

Some of basic theorems- High power transfer and Thevenin's theorem- R, C, L circuits, filters and resonance- Current and voltage difference distribution - Equivalent electric circuits for bio membranes - Dielectric constant and bio applications - P-N junction - current rectifier - Transistors and current amplification current for biological applications.

References:

- "Integrated electronics: analog and digital circuits and systems" By: Jacob Millman, McGraw-Hill Inc. Japan.
- "Applied electricity for engineers" By: Boris Kuznetsov, Mir Publishers-Moscow.

PHYS373 Molecular Spectroscopy and its Bio Applications:

2 Credit hrs. Lecture + 1 Credit hr. Practical

Prerequisites Courses: PHYS231

Chemical systems: Atom – Molecules – Ultraviolet and visible absorption spectroscopy UV and visible absorption instrumentation (Radiation source wavelength control Sample containers – Detector – Recording or meter) – Applications of ultraviolet and visible absorption spectroscopy – Absorbing species, Absorption by organic compounds, absorbing by inorganic compounds, qualitative and quantitative analysis.

Infrared absorption spectroscopy – Theory of infrared absorption - Infrared instrumentation – Sample handling – Application of infrared absorption spectroscopy.

Magnetic resonance spectroscopy – Nuclear magnetic resonance (NMR) – Principle of NMR spectroscopy nuclear magnetic resonance spectrometer (the magnet – magnetic field sweep generator – Radiofrequency) – The absorption detector sample holder and sample probe sample preparation – Chemical shift (δ) – Spin-spin coupling identification of molecular structure by NMR imagine.

References:

- "Fundamentals of Optics" By: H.White & F.Jenkins

PHYS374 Replacement of Vital:

2 Credit hrs. Lecture + 1 Credit hr. Practical

Prerequisites Courses: -

Materials used in the alternative: Metals- Ceramics- Polymers hint of their Profile and chemical properties: The electrical characteristics of the materials vital alternative: a study of electrical properties in the field of static (Relations of electrostatic - Molecular ionization - and the relationship of the local area and Klacjus Musety - A consistent theory of isolation and molecular weight).General theory of relaxation. Different ways of measuring the electrical properties (such as fixed unarmed - and other types of resistance) of alternative materials. Mechanical properties of materials vital alternative. The basic properties of the materials used to measure hardness- fracture- fatigue- stress corrosion with the properties of the materials used in alternative materials. Response and body tissues after the replacement of the joints and organs- The interaction of bone with its alternative study how to form the roof line between the natural bone materials and install a replacement- The cause of the collapse of the stabilization alternative. Applications for metals as an alternative body interactions with metals and polymers and ceramics.

References:

- "Dielectric Behavior of Biological Molecules in Solution" By: Grant, R.J. Sheppard and G.P. South.

PHYS375 Quantum Mechanics and Biological Application:

2 Credit hrs. Lecture

Prerequisites Courses: PHYS241

Modern Quantum Theory – Assumption in Quantum Theory – Operators – Hermethien Operators – Schrodinger Equations in one and three dimensions – Properties of

Schrodinger's Equation – Central Potential and Hydrogen Atoms - Applications on Quantum Mechanics (Infinite Well) – (Quantum Well). Scattering of particles from Potential Barrier - Scattering of particles from finite potential barrier – Evaluation of electronic energy level of diatomic molecule - Calculation of vibrational energy level for diatomic molecule – Study of rotational energy level for diatomic molecule.

References:

- "Introduction to Quantum Mechanics" By: Dick & Wittke.
- "Quantum Physics" By: Gasiorowicz.

PHYS376 Radio-Biophysics:

2 Credit hrs. Lecture + 1 Credit hr. Practical

Prerequisites Courses: PHYS331

Atomic nucleus – Isotopes – Atomic mass unit – Elementary particles in radiotherapy – Natural radioactivity and radioactive series – Nuclear decay laws – Total exposure dose – Nuclear decay schemes – Radiation energy absorption – Population of radioactive decay products – Short lived progenies – Radioactive isotope production – Interaction of different kinds of radiation with biological tissues – Charged particle ranges in different media – Interaction of gamma rays with matter – Ionizing radiation detection.

References:

- "Concepts of Radiation Dosimetry" By: Kase & Nelson.
- "Medical Radiation Physics" By: Hendee.

PHYS377 Electronic Simulation of Bio Systems:

2 Credit hrs. Lecture + 1 Credit hr. Practical

Prerequisites Courses: PHYS251

Electronic operational amplifiers- uni-input circuit: voltage adder- current to voltage converter- integrator, differentiator and summing- Bi-input circuit: subtractor circuit- differential integrator- voltage follower - Non-linear circuits: comparator- electronic operational amplifier in simulation- Hook law; spiral and mass system –Analog calculus - Differential equations - Heart pulses simulator –Heart pulse filters – Filter oscillators - Listening evaluation simulating rhythm of adjusting nerve -Detector skin resistance galvanometer.

References:

- "Integrated electronics: analog and digital circuits and systems" By: Jacob Millman, Mc Graw-Hill Inc. Japan.

PHYS378 Mathematical Biophysics:

2 Credit hrs. Lecture

Prerequisites Courses: PHYS172

Study of various bio physics model – Model of single and two rooms – Model depends on differential equations – Model of enzyme motion – Model fitting with experimental data – Mont carlo simulation – Bio physics mont carlo application.

References:

- "Mathematical Methods for Physicists" By: Arfke.
- "Mathematical Methods in Physics Science" BY: Boas .

PHYS471 Bio-Mechanics:

2 Credit hrs. Lecture + 1 Credit hr. Practical

Prerequisites Courses: PHYS375

Study of the Bio mechanics for bones- Flow equation- Elasticity for the rigid bodies- Velocity of liquids- Elasticity and Viscosity for the elasticity viscous materials and its properties- Study and exam the mechanical properties for the elasticity viscous materials- The blood flow- Flow of the blood from the large to small intensity.

References:

- "Biophysics" By: Elsayed, Shamloula and Ebeid.
- "Aspects of Biophysics" By: William Hughes.

PHYS472 Laser's Physics and its Bio Applications:

2 Credit hrs. Lecture + 1 Credit hr. Practical

Prerequisites Courses: PHYS373

Laser beam – Types of lasers – Interact of laser beam with tissues – Sources of laser beam – Safety and dangerous of laser beam – Biological effects of laser beam – Methods of laser beam measurement – Applications of laser beam in medicine, chemistry and bio science – Evaluation of laser beam dangerous.

References:

- "Physical Principles of Ultrasonic Diagnosis " By: P.N.T. Wells.
- "Concepts of Radiation Dosimetry" By: Kase & Nelson.
- "Medical Radiation Physics" By: Hendee.

PHYS473 Biophysics of The Nervous System:

2 Credit hrs. Lecture + 1 Credit hr. Practical

Prerequisites Courses: PHYS372

Introduction- The composition and function of the nervous system- Physics is vital for nerve cells- Is a cell membrane- Moving chemical- Electric switch- Physics is vital for ion channels- Diverse ways of measuring the activity of nerve devices- A comprehensive theory- Sensory organs, neural networks- Learning and memory- Devices used in measuring the electrical activity of nerve cells- Registration inside and outside the cells- Control signals- poles- Error measurements methods- the analysis results.

References:

- "Biophysics" By: Elsayed, Shamloula and Ebeid.
- "Fundamentals of Physics " By: HallidayResnick.

PHYS474 Conformal Radiotherapy:

2 Credit hrs. Lecture

Prerequisites Courses: PHYS376

Radiation dose units of measurement – Exposure and absorbed dose – Distribution of equal doses in tissue equivalents – Types of tissue equivalents – Radiation square and circular fields equivalent to rectangle fields – Percentage dose and back scattering – Tissue weighting factors – Radiation weighting factor – Dose tissue to air ratio – Dose tissue to tissue equivalent ratio – Patient body curvature effect on dose contours and determination – Tissue in homogeneity and its effect on dose contours – Wedge high energy degraders – Accumulated dose – Dose contours for different dual and triad mixed radiation fields – Manual addition of radiation fields – Radiation field directing devices – Determination of therapeutic radiation dose as a function of the percentage dose and Source-to-target distance – Receipt of determined therapeutic radiation dose – Determination of therapeutic radiation dose in terms of air percentage dose and source to pivot distance in target tissue – Correlation between percentage radiation dose and relative tissue to air dose.

References:

- "Medical Radiation Physics" By: Hendee.
- "Radiation Dosimetry" By: Attix & Roesch Tochilin.

PHYS475 Contact Cell Biophysics:

2 Credit hrs. Lecture + 1 Credit hr. Practical

Prerequisites Courses: PHYS377

Overview of the cell signals-signals the cell fertilized during the early history of life-connected cells can be close or far apart- the three phases of the cell signals- reception-conversion- response- the reception of signal is to start response- the issues of The Chemical protein cause future change in the form of protein. The receptors are most of the plasma membrane proteins. The transfer corridors, the corridors carrying a signal from the receptors to the cell receptors. Phasara proteins are the general method of regulation within the cell and transfer the predominantly technical reference.

References:

- "Biophysics" By:Elsayed, Shamloula and Ebeid.
- "Aspects of Biophysics" By: William Hughes.

PHYS476 Biophysics of Membranes and Cell:

2 Credit hrs. Lecture + 1 Credit hr. Practical

Prerequisites Courses: BOT332

The importance of biological tissue- To protect the cell- With optional communication Center- storage of materials inside the booths inside the living transport system Eng chemical- Biological membrane acquaintance. The molecular structure of membranes- biological activities: a model of fluid- dynamic model Almoszek is the biological membrane model- the models manufactured for the biology membrane lab- fat single membrane on the surface of water air- Allibozzomat fat black film, permeability of membranes and movement- The impact of both the temperature effect on the permeability effort nicknamed Permeability channels within the cell membrane.

References:

- "Dielectric Behavior of BiologicalMolecules in Solution" By: Grant, R.J. Sheppard and G.P. Sou.
- "Biophysics" By: Elsayed, Shamloula and Ebeid.

PHYS477 Molecular Biophysics:

2 Credit hrs. Lecture + 1 Credit hr. Practical

Prerequisites Courses: PHYS375

Energies and statistics in the living cell: Dynamics- Statistical and biological science - the theory of absolute rates of interactions - Roads have to be physical size and form of

particles: Random movement and proliferation - Spreading factor measurement- Sedimentation method Poise have to be a way n factor deposition speed to be settling factor - Factor rotational spread and Refractive bilateral - Molecules strong vitality - The neighborhood chemistry - Transport active tissue Models -Photosynthesis - Transpiration

References:

- "Aspects of Biophysics" By: William Hughes.
- "Dielectric Behavior of Biological Molecules in Solution" By: Grant, R.J. Sheppard and G.P. South

PHYS478 Biophysics Health:

2 Credit hrs. Lecture

Prerequisites Courses: PHYS376

Ionizing radiation: international bodies and international organizations interested in measurement of ionizing radiation. Dosimetry standards. Room ionization (open air absolute) standard. Measuring the radiation absorbed dose- kinetic energy flowing in the center Karma. Gamma radiation of the qualitative. Account absorbed doses of radioactive elements within the body of a beta or gamma radiation. Principal rules for the prevention of ionizing radiation- exposure of individuals in acts of radiation and the general public. Instrumentation doses personal scale enclave- the film badge- the measure doses fluorescence gases. Non-ionizing radiation: deal-ray laser tissues- sources and types of lasers- safety rules and the risks of lasers- vital to the effects lasers- laser measurement methods- standards protective- risk assessment and laser- some uses of lasers in medicine- chemistry and biosciences.

References:

- "Laser in Biology and Medicine" By: F. Hillenkamp, R. Partesi C.A. Sacc.
- "Radiation Dosimetry" By: Attix & Roesch Tochilin.

PHYS103 Basics of Computer Science (1):

2 Credit hrs. Lecture + 1 Credit hr. Practical

Prerequisites Courses: -

The design and functioning of hardware - computer structure - the fundamental terminology associated with computers – Deal with the well-known operating system – word – excel - power point and using internet in academic search.

PHYS104 Basics of Computer Science (2):

2 Credit hrs. Lecture + 1 Credit hr. Practical

Prerequisites Courses: -

Fortran definition - Simple FORTRAN 90 program - The numerical constant and variable-Mathematical expressions - Input and output sentences - The conditional sentences - the conditional loop - The logical expressions and the control sentences - Function principle and subroutines - Matrix definition - Form description or primary coordination - Letters and number implementation-program examples - Matrix methods - Evidence of some kinds - Advanced applications - Files implementation - Kind of statements and applications.

References:

- Stephen j.Chapman,"Fortran 90/95 for scientists and engineers, Mc Graw Hill 1998,ISBN 0070119384.

PHYS261 Computer Architecture and Organization:

2 Credit hrs. Lecture + 1 Credit hr. Practical

Prerequisites Courses: -

The basis of organization and designing - Input - Output and end - Computational programming of the arithmetical and logical processes - Input and output programming - Data transfer - Control memory - Designing algorithmic and arithmetical treatment - Non simultaneous database - Direct input memory - The hypothetical memory - Various treatments.

References:

- John L. Hennessy and David A. Patterson (2003). Computer Architecture: A Quantitative Approach, Third Edition, Morgan Kaufmann Publishers, Inc. ISBN 1558605967.

PHYS262 Fundamentals of Structure Programming:

2 Credit hrs. Lecture + 1 Credit hr. Practical

Prerequisites Courses: -

Constant and variables - Mathematical operators - Logical operators and relationships - complex operators - Input, output sentences and data forms - Comment sentences - Control structure - Repetitions sentences - Subroutines - Functions and implementation - matrices - Records - files - Repetition function - programming language.

References:

- Stroustrup, Bjarne (2000). The C++ Programming Language, Special Edition, Addison-Wesley. ISBN 0201700735.

PHYS263 Data Structures and Algorithms:

2 Credit hrs. Lecture + 1 Credit hr. Practical

Prerequisites Courses: PHYS262

Abstract Data Types (ADT). Stacks: Definition and operations - Implementation of stacks with array and records - Applications of stacks. Queues: Definitions - Implementation of queues - Applications of queues. Linked lists: Singly linked lists - linked stacks - Linked queues - Doubly linked lists - Application of linked lists. Tree structures - binary trees: Binary tree traversals - Binary tree search. Searching algorithms - Sequential search. Sorting algorithms: insertion sort- Selection sort.

References:

- Stroustrup, Bjarne (2000). The C++ Programming Language, Special Edition, Addison-Wesley. ISBN 0201700735.

PHYS264 Object Oriented Programming:

2 Credit hrs. Lecture + 1 Credit hr. Practical

Prerequisites Courses: PHYS262

Objects - Object classes - Deriving an object oriented design. Functional oriented design: Data flow diagrams - Structure charts - Inheritance - A design example - Data dictionaries - Deriving structure charts - Design examples - Concurrent systems design. User interface design: User interface design objectives - Interface metaphors - WIMP (Window, Icons, Menus, and Pointing) interfaces using color displays.

References:

- Schach, Stephen (2006). "Object-Oriented and Classical Software Engineering", Seventh Edition. McGraw-Hill 2006. ISBN 0073191264.

PHYS265 File Organization and Processing:

2 Credit hrs. Lecture + 1 Credit hr. Practical

Prerequisites Courses: PHYS262

File introduction - General view for filing - Sorting and external merging algorithm - Random access - Perfect scattering - File organization in structured tree - Multi key filing system - Invert files.

References:

- Alan L. Tarb . "File organization and processing", by Maloy lithographic 2007, ISBN 0471605212.

PHYS361 Database management system:

2 Credit hrs. Lecture + 1 Credit hr. Practical

Prerequisites Courses: PHYS263

Overview of database administration - Data and autonomy - Building of database systems - Administration system of the database - Overview of DB2 - SQL language- Arithmetic and algebra relationships - Data definition - Viewing definition - Logical relationships of data.

References:

- Elmasri Navathe." Fundamentals of Database Systems", ISBN 0805317554, Addison Weseley 2000.

PHYS362 Image Processing (1):

2 Credit hrs. Lecture + 1 Credit hr. Practical

Prerequisites Courses: PHYS264

The origins of digital image processing – examples of fields that use digital image processing- fundamental steps in digital image processing – components of an image processing system – elements of visual perception – light and the electromagnetic spectrum – image sensing and acquisition – image sampling and quantization – some basic relationships between pixels – linear and nonlinear operations – image enhancement in the spatial domain.

References:

- Rafael C.Gonzalez, Richard E.Woods, "digital image processing", India, 2002.

PHYS363 Computer Network:

2 Credit hrs. Lecture + 1 Credit hr. Practical

Prerequisites Courses: -

The use of computer networks - Network architecture - the ISO reference model, examples of LAN networks. Network topology - TCP/IP - backbone design - local access network design. Data transmission - Transmission impairments - Channel capacity - Transmission media - Guided transmission - Unguided transmission - Multiplexing, Data encoding - The data link layer - Elementary data link protocols-flow control - Error detection- analysis of protocols - Wide area networks - packets and datagrams - Routing algorithms - flooding-satellite packet broadcasting. Local area networks - Examples of LAN networks - Ring LAN networks - Star LAN-Wireless networks - File transfer protocols - Distributed data base systems - distributed computations.

References:

- William Staillings . “Data and Computer Communication”, ISBN 0130843709, Printic Hall 2001.

PHYS364 Artificial Intelligence (1):

2 Credit hrs. Lecture + 1 Credit hr. Practical

Prerequisites Courses: PHYS264

Introduction to artificial intelligence concepts and definitions - knowledge acquisition and representation: Predicate Calculus - Structured Representations and Network Representations - State Space Search: trees and graphs. Heuristic search - model based reasoning - case-based reasoning, reasoning with uncertain or incomplete knowledge - Overview of AI languages: database - Overview of AI Application Areas - expert systems - artificial intelligence computers-relationship between the artificial intelligence and the synthetic nervous nets.

References:

- Floriana Esposito, Zbigniew W. Ras, Donato Malerba, "Foundations of Intelligent Systems: 16th International Symposium, ISMIS 2006", Italy, 2006, (Lecture Notes in Artificial Intelligence). ISBN: 354045764X.

- Ivan Bratko, "Prolog Programming for Artificial intelligent", ISBN 0201403757, Addison-Wesley 2001.

PHYS365 Computer Graphics:

2 Credit hrs. Lecture + 1 Credit hr. Practical

Prerequisites Courses: PHYS264

Introduction to computer Graphics - Overview of Graphics systems - Line drawing algorithms - Circle drawing algorithms - Ellipse drawing algorithms - Area filling algorithms - polygon filling algorithms - line clipping algorithms - Polygon clipping algorithms - Two dimensional transformations - (translation-rotation-scaling) general transformations-composite transformations); Three dimensional object representation and projections - three dimensional modeling and transformations(translation-rotation-scaling-shear-reflection-composite) - Three dimensional Viewing and Camera Model.

References:

Bradley Kjell, "Vector Math for 3D Computer Graphics", 2003.

PHYS366 Algorithms design and analysis:

2 Credit hrs. Lecture + 1 Credit hr. Practical

Prerequisites Courses: PHYS263

Overview of designing - analysis and complexes algorithm - General method - Double search method - Arrangement by choice - Shorter generated trees - Dynamic programming - Attributes method for searching in the trees - Method of the pursuit for the back - Octagonal files problem - Koke's theory.

References:

- Gilles Brassard and Paul Bratelly, "Algorithms theory and practice", prantic hall international, 1988, ISBN 013023169X.

PHYS367 System Design and Analysis:

2 Credit hrs. Lecture + 1 Credit hr. Practical

Prerequisites Courses: PHYS263

Essential concepts - Definition (system - employee) - System analyzed- Data dictionary - English language assembly - Decision tree - Study of data - Design ways - The physical component for system design - Programming system - Files system - Determination of logical operations - Design and control program - Files and reports.

Reference:

- DeMarco, Tom. "Structured Analysis and System Specification" ISBN 0138543801

PHYS461 Modelling and Simulation:

2 Credit hrs. Lecture + 1 Credit hr. Practical

Prerequisites Courses: PHYS263

Basic simulation modeling. Nature of simulation models - Building models & simulation
Simulation output data and stochastic processes - Validation of simulation models -
Verification of simulation computer programs. Types of simulation models: discrete event
simulation, continuous simulation - Application in mathematical models - Stochastic
simulation - application in stochastic simulation Simulation of a single-server queuing
system - Simulation of random walk - Simulation of traffic movement - Statistical
procedures for computing real-world observation & simulation output data - Some
practical considerations: Probability distributions - Random number generators -
Generating random variables using of simulation software packages.

References:

- Roger McHaney, "Computer Simulation A practical Perspective", Academic Press, 1999.
- Hamdy A. Taha, "Operations Research", Collier Macmillan Publishers, 1989.

PHYS462 Image Processing (2):

2 Credit hrs. Lecture + 1 Credit hr. Practical

Prerequisites Courses: PHYS362

Image enhancement in the frequency domain – introduction to the Fourier transform and
the frequency domain – smoothing frequency domain filters – sharpening frequency
domain filters – image restoration – noise models – geometric transformation – color
image processing – color fundamentals – wavelets processing – image compression –
image segmentation.

References:

- Rafael C. Gonzalez, Richard E. Woods, "digital image processing", India, 2002.

PHYS463 Data Mining:

2 Credit hrs. Lecture + 1 Credit hr. Practical

Prerequisites Courses: PHYS361

data mining concept - Inductive learning, Machine Learning - Differences between Data Mining and Machine Learning - Data Mining Models - Characteristics of a data warehouse - warehouse model - data mine problems - data mine applications - Data Mining Functions - Classification - Associations - Sequential/Temporal patterns - Clustering/Segmentation - Data Mining Techniques - Cluster Analysis - induction - deduction - decision tree - neural network - data visualization - data mining examples.

References:

- Morgan Kaufmann, "Data Mining: Concepts and Techniques", March 2006.

PHYS464 Wireless Network:

2 Credit hrs. Lecture + 1 Credit hr. Practical

Prerequisites Courses: PHYS363

• Radio frequency spectrum. How simple wireless networks work - How data is transmitted by waves - Antenna - Propagation modes - Satellite communication - Radio and television broadcasting - Cellular networks how cell phone connect the network, handoff - Power control - Cellular channel - Analog and digital mobile systems - TDMA - CDMA and GSM wireless LAN technology - Blue tooth.

References:

- William Stallings. "Wireless Network and Communication", ISBN 0130843709, Prentice Hall 2005.

PHYS465 Artificial Intelligence (2):

2 Credit hrs. Lecture + 1 Credit hr. Practical

Prerequisites Courses: PHYS364

Pattern recognition systems – sensing – segmentation and grouping – feature extraction – classification – the design cycle – data collection – feature choice – training – supervised learning – unsupervised learning – Bayesian decision theory – minimum error rate classification – Bayes decision theory – missing and noisy feature – Bayesian estimation – Bayesian parameter estimation – multilayer neural networks.

References:

- Richard O. Duda, et al., Pattern Classification, Canada, 2001.

PHYS466 Computer Security:

2 Credit hrs. Lecture + 1 Credit hr. Practical

Prerequisites Courses: PHYS363

Characteristics of computer intrusion - Points of security vulnerability - Methods of defense. Basic encryption and decryption, Secure encryption systems. The data encryption standard (DES) - Enhancing cryptographic security. Security involving programs: Information accesses problems - Viruses and worms - Controls against attack - Operating system control. Design of secure operating systems: Models of security - penetration of operating systems - Database security - Computer network security and communication security.

References:

- Deborah Russell, and G.T. Gangemi sr ,” Computer Security Basics “,Oreilly 2006, ISBN 0596006691.

PHYS467 Multimedia:

2 Credit hrs. Lecture + 1 Credit hr. Practical

Prerequisites Courses: PHYS362

Multimedia Techniques - Screen resolution and screen technology - Video accelerator design system - Raster graphics (3D-transformation) - Analog-to-digital conversion - Video compression - Mixing and displaying at 30 FPS with full color capacity. Physics of Sound - Sound cards - Sound cards limitations - Mixing sound video and voice traffic control - Animation.

References:

- Multimedia Making it Work, by Tay Vaughan. Osborne McGraw Hill, 1993. ISBN 007881869.

PHYS468 Electronic Learning and digital Library:

2 Credit hrs. Lecture + 1 Credit hr. Practical

Prerequisites Courses: PHYS467

Using of technology in learning - Distant learning - Site design - What is the digital library - Geographic region - Digital library and information community - Interaction and intercession integrity in digital library - Electronic publishing.

References:

- Greenstein, Daniel I., Thorin, Suzanne Elizabeth. The Digital Library: A Biography. Digital Library Federation (2002) ISBN 1933645180.

PHYS469 Selected Course in Computer:

2 Credit hrs. Lecture + 1 Credit hr. Practical

Prerequisites Courses: -

Course which are not included in the curriculum and seems to be needed should be suggested as an elective course by department.